MILITARY STANDARD

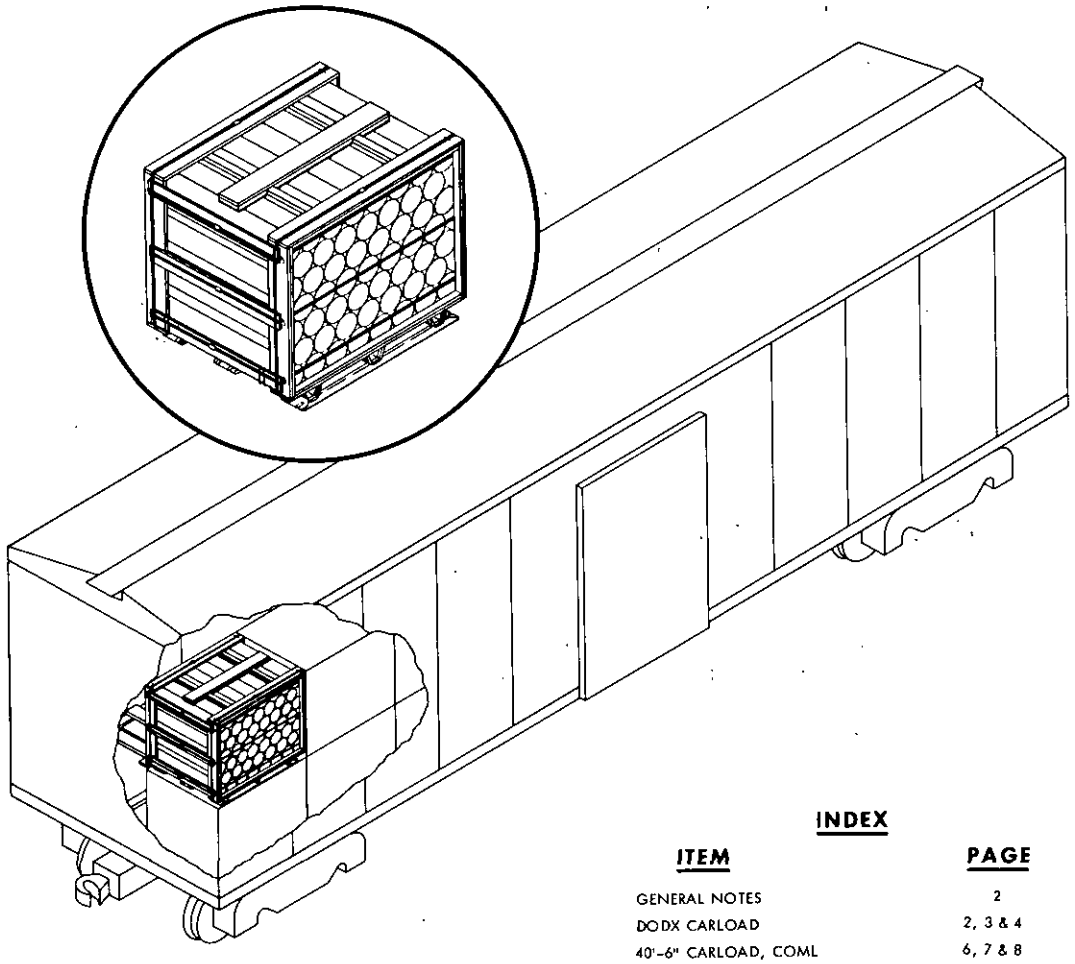
MIL-STD-1325-152
(NAVY)
17 JULY 1973

CARLOADING

CARTRIDGE, 3"/50 IN TANK MK 5 AND MODS
DOMESTIC UNIT LOAD

UNIT LOAD DATA

UNIT LOAD DRAWING . . . . . WR-53 /774
WEIGHT . . . . . 1845 LBS
DIMENSIONS . . . . . 51 1/2 L x 40 1/2 W x 43 3/4 H
CUBE . . . . . 52.8 CU FT
DOT HAZARD CLASSIFICATION. . . . . EXPLOSIVES A



INDEX

ITEM	PAGE
GENERAL NOTES	2
DODX CARLOAD	2, 3 & 4
40'-6" CARLOAD, COML	6, 7 & 8
50'-6" CARLOAD, COML	10, 11 & 12

NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCK-LOADING AND CONTAINERLOADING MILITARY STANDARDS, REFER TO INDEX TO STANDARDS MIL-HDBK-236 (NAVY).

AUTHORIZED AND RELEASED FOR GENERAL USE

W. S. Gray NWHL 7/3/73
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J. E. Kelly 7/3/73
SIGNATURE (ORD) SYSCOM, BY DIRECTION DATE

APPROVED BY BUREAU OF EXPLOSIVES

A. F. Grassmuck MILITARY ASSISTANT 5/2/73
SIGNATURE DATE

ORIGINATOR L. R. Cairys 7/2/73
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. FOR GENERAL INFORMATION CONCERNING ORDERING, INSPECTING, AND PREPARING CARS, AND FOR DUNNAGING MATERIALS, DESIGN, AND INSTALLATION OF DUNNAGE SEE THE GENERAL DOCUMENT MIL-STD-1325 (NAVY) "RAILCAR LOADING OF HAZARDOUS MATERIALS."
2. WHEN PLANNING SHIPMENTS ORDER THE MINIMUM NUMBER OF CARS OF THE CAPACITY REQUIRED FOR THE SHIPMENT. UTILITY LOADER CARS SHALL BE SERIES DODX 28000.
3. LOADING PLANS SHOWN ARE FOR DODX UTILITY LOADER CAR WITH 50 FT 6 INCHES INSIDE LENGTH, 107 3/4 INCHES INSIDE WIDTH BETWEEN RAILS (111 INCHES INSIDE WIDTH BETWEEN SIDE WALLS), COMMERCIAL BOXCARS WITH 40 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH, AND COMMERCIAL BOXCARS WITH 50 FT 6 INCHES INSIDE LENGTH, 110 INCHES INSIDE WIDTH.
4. THE "LOAD LIMIT" OF A CAR MUST NOT BE EXCEEDED NOR SHOULD THE RAILCAR BE LOADED SO THAT MORE THAN ONE-HALF OF THE *"LOAD LIMIT" IS CARRIED BY ONE SET OF TRUCKS.*
5. IF END WALLS OF CARS ARE NOT SQUARE THEY MUST BE SQUARED OFF BEFORE STARTING TO LOAD CAR.
6. THE LOAD CONSISTS OF 3"/50 CARTRIDGES PALLETIZED IN ACCORDANCE WITH WR-53/774.
7. THE UNIT LOADS ARE HANDLED AND LOADED WITH A SUITABLE FORK LIFT TRUCK.
8. UNLESS OTHERWISE SPECIFIED NAILING SHALL BE IN ACCORDANCE WITH MIL-STD-1325 (NAVY).
9. ALL STRAP JOINTS SHALL BE CRIMP-TYPE JOINTS MADE WITH A CRIMPING TOOL THAT DOES NOT CUT THE SEAL OR STRAP, RATHER THAN A TOOL THAT MAKES A NOTCH-TYPE JOINT. THE STRAPPING MUST BE IN ACCORDANCE WITH NAVSUP NOTICE 4642 DATED 21 APRIL 1971.
10. APPLICABLE MATERIAL SPECIFICATIONS:

    DUNNAGE LUMBER - FED. SPEC. MM-L-751

    NAILS - FED. SPEC. FF-N-105

    STRAPPING - FED. SPEC. QQ-S-781, TYPE I, HEAVY DUTY, CLASS A, DRY (UNLUBRICATED)

    SEALS - FED. SPEC. QQ-S-781, STYLE III, HEAVY DUTY
11. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED ATTACH SHIPPING DOCUMENTS INSIDE THE CAR IN AN ACCESSIBLE AREA, CLOSE AND SEAL BOXCAR DOORS, AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF CAR AS PRESCRIBED IN OP 2165 (VOL 1).

50 FT 6 IN. BOXCAR, DODX

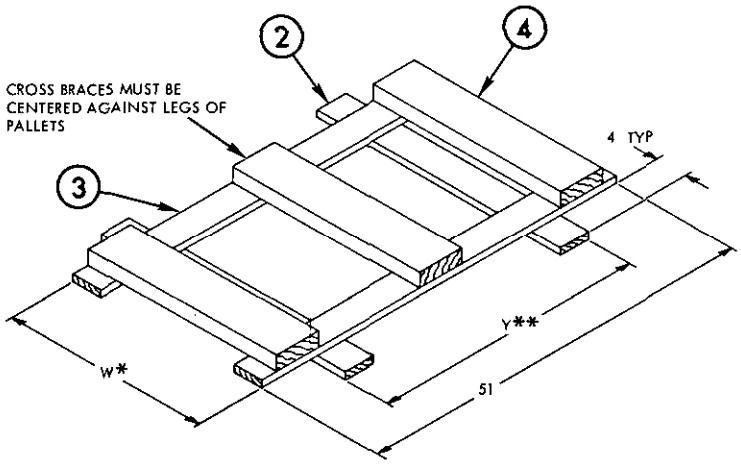
1. THE CARLOAD CONSISTS OF 40 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. A DETAILED DESCRIPTION AND OPERATING INSTRUCTIONS FOR THE UTILITY LOADER ARE CONTAINED IN OP 1750.
3. TO PREVENT UNUSED "DF" EQUIPMENT FROM BECOMING DISLODGED DURING TRANSIT OF DODX CARS SECURE IT AT ANY LOCATION IN THE BOXCAR WHICH WILL NOT INTERFERE WITH UNLOADING.
4. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN DODX BOXCARS THE SAME PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE. ANY BAYS OR PORTION THEREOF MAY BE USED PROVIDING THE WEIGHT DISTRIBUTION REQUIREMENTS OF THE AAR ARE COMPLIED WITH, SEE MIL-STD-1325 (NAVY). EACH CROSSMEMBER WILL BE USED IN SUCH A MANNER THAT IT WILL RETAIN NOT MORE THAN 2000 LBS OF THE LADING.

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

4	CROSS BRACE	2 x 6 x W*	30	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 51	20	2, 4	SEE NOTE **	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	20	SEE 3	-	-
1	SLEEPER	2 x 4 x 51	20	CAR FLOOR	2 PER FOOT	16d
PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
LIST OF MATERIALS & NAILING DATA						

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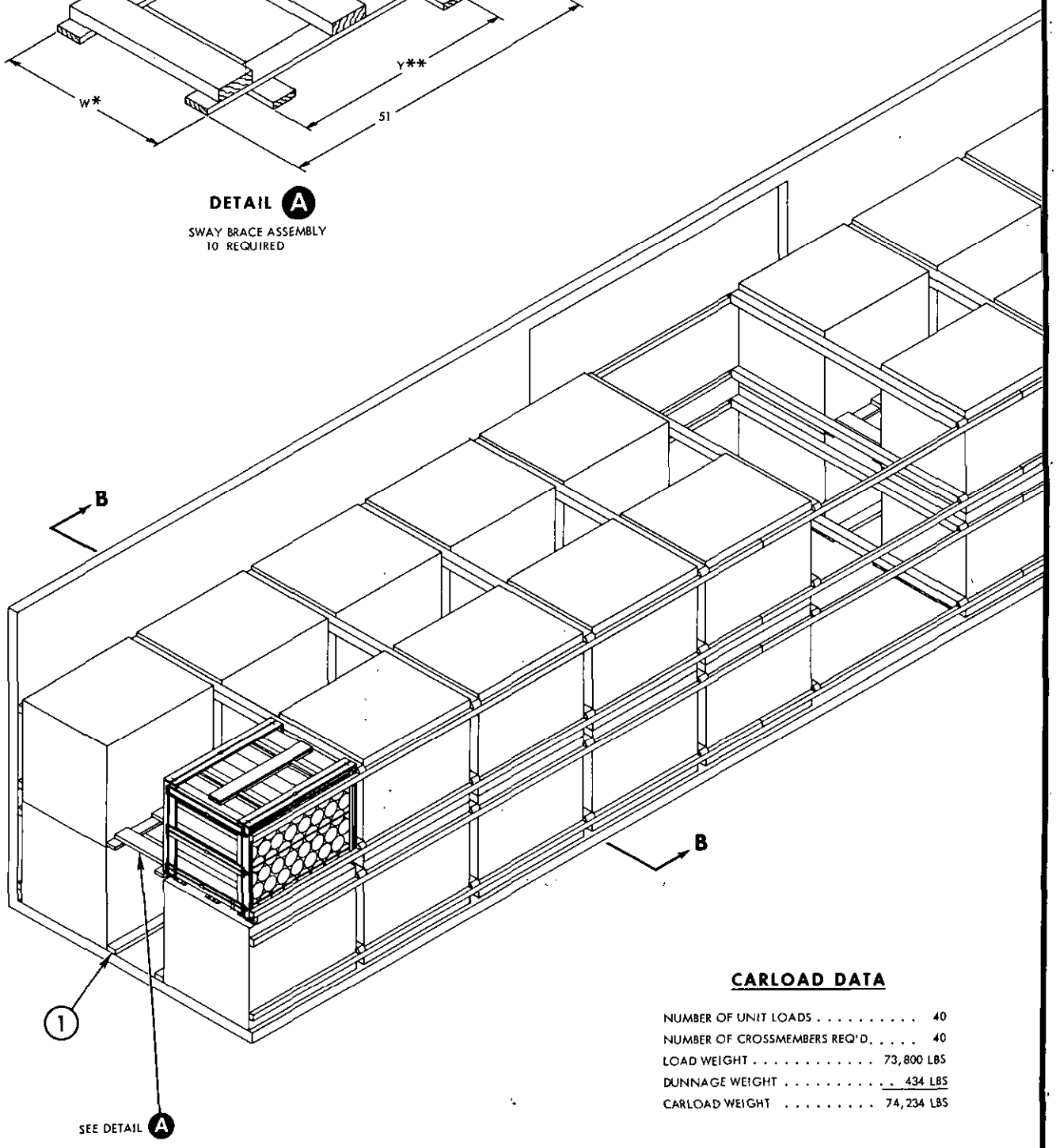
DETAIL A

SWAY BRACE ASSEMBLY
10 REQUIRED

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

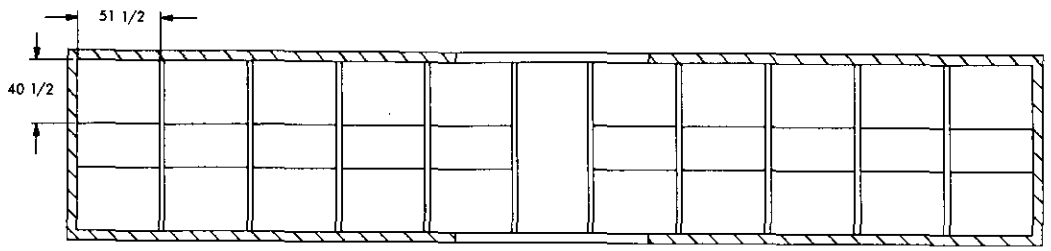
INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS



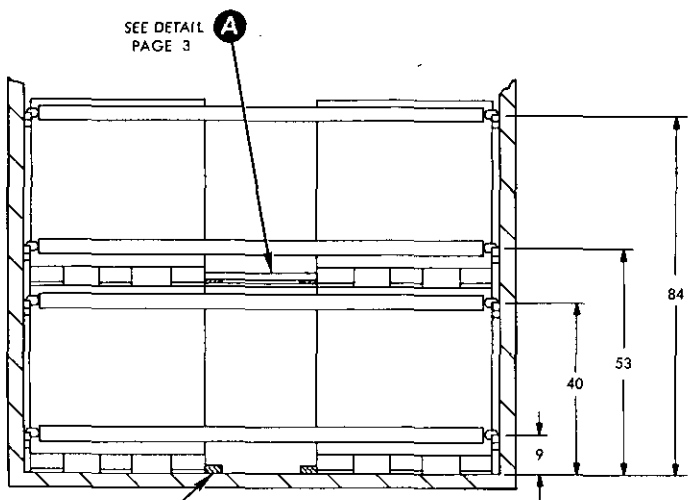
CARLOAD DATA

NUMBER OF UNIT LOADS	40
NUMBER OF CROSSMEMBERS REQ'D.	40
LOAD WEIGHT	73,800 LBS
DUNNAGE WEIGHT	434 LBS
CARLOAD WEIGHT	74,234 LBS

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LOADING PLAN



SECTION B-B

SHOWING LOCATION OF WALL MEMBERS AND DOORWAY MEMBERS USED FOR CROSSMEMBERS

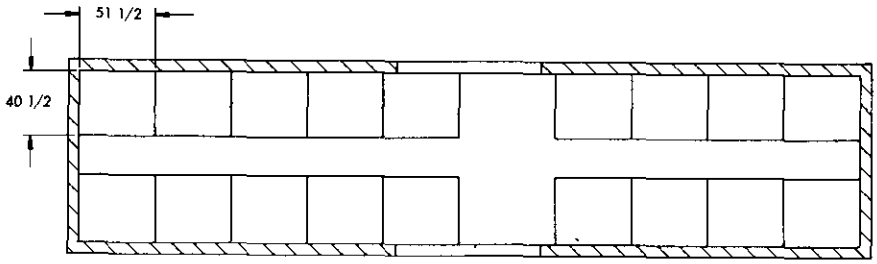
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40 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 36 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 6 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 6 FT WIDE.



LOADING PLAN

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

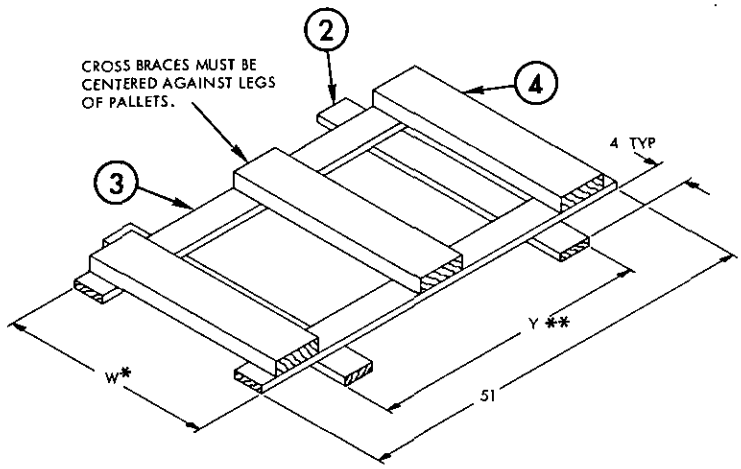
**PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

***2 x 6 STRUTS DOUBLED AND LAMINATED WITH 10d NAILS MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

PIECE NO.	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NAILS NUMBER	NAILS SIZE
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 96	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 x 6 x 48	2	9	2 PER FOOT	10d
9	GATE HOLD DOWN	2 x 6 x 10 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 90	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	27	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 51	18	2, 4	SEE NOTE**	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	18	SEE 3	-	-
1	SLEEPER	2 x 4 x 51	18	CAR FLOOR	2 PER FOOT	16d

LIST OF MATERIALS & NAILING DATA

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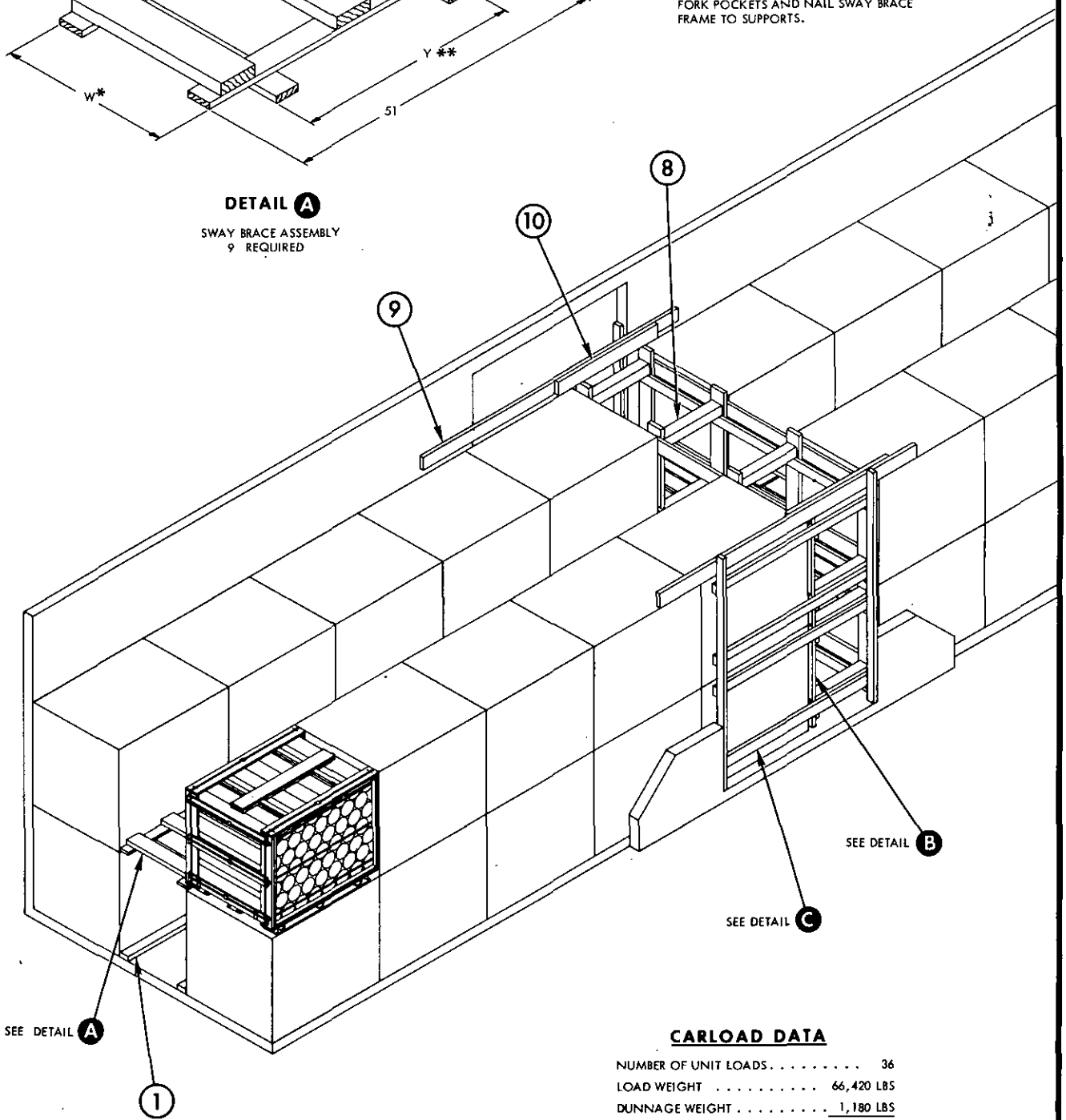


*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.
**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

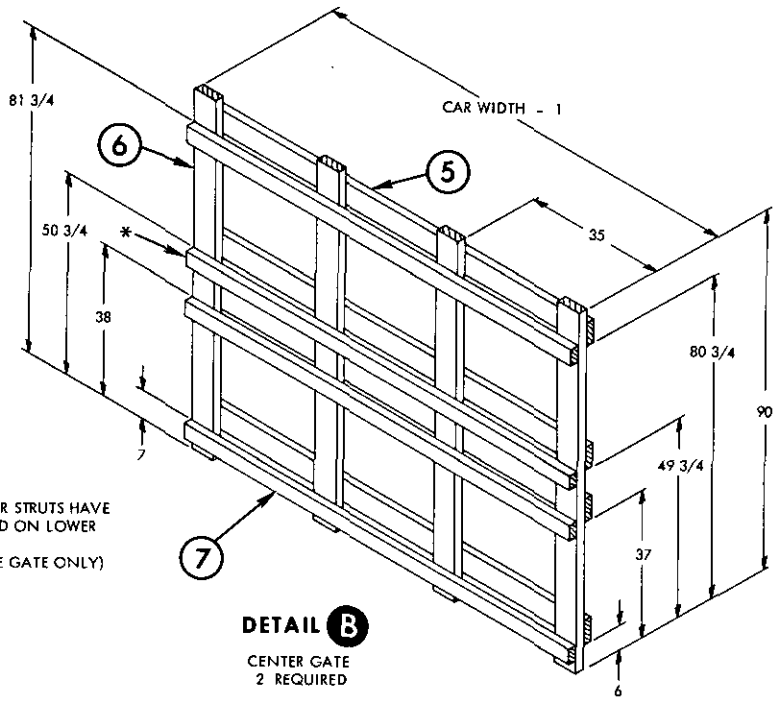
SWAY BRACE ASSEMBLY
9 REQUIRED



CARLOAD DATA

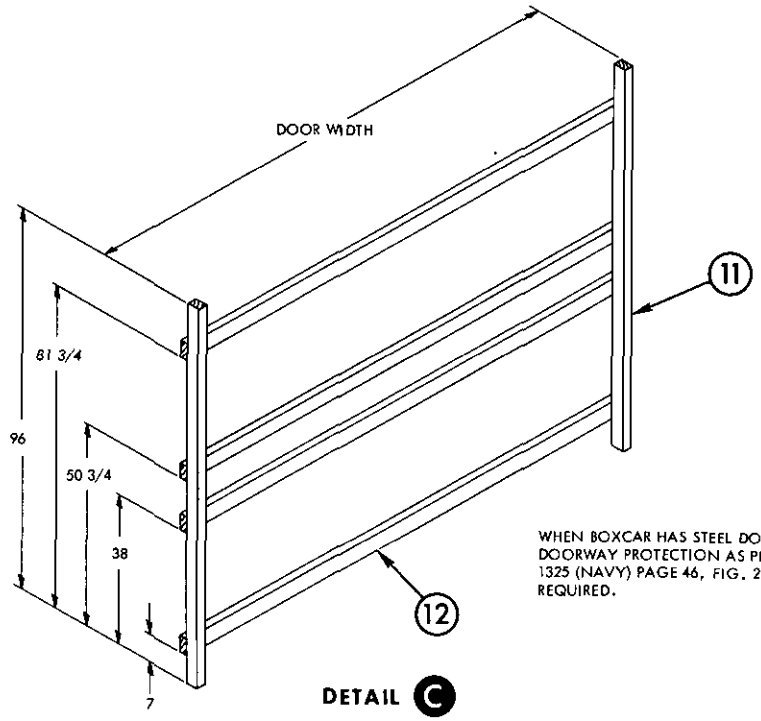
NUMBER OF UNIT LOADS	36
LOAD WEIGHT	66,420 LBS
DUNNAGE WEIGHT	1,180 LBS
CARLOAD WEIGHT	67,600 LBS

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*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)

DETAIL B
CENTER GATE
2 REQUIRED



WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.

DETAIL C
DOORWAY PROTECTION
2 REQUIRED

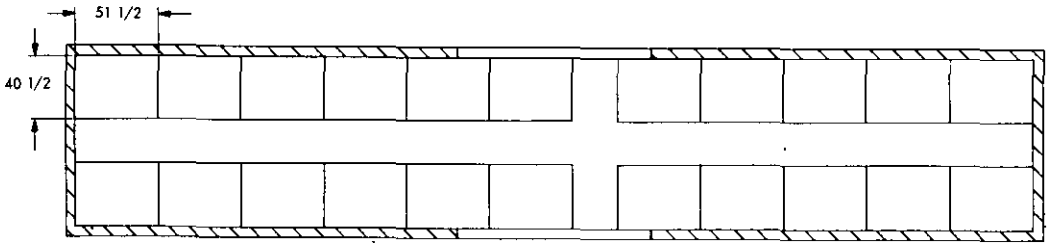
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50 FT 6 IN. BOXCAR, COMMERCIAL

1. THE CARLOAD CONSISTS OF 44 UNIT LOADS WHICH MUST BE LOADED AND DUNNAGED IN ACCORDANCE WITH THIS PROCEDURAL DRAWING.
2. WHEN LESS THAN CARLOAD (LCL) QUANTITIES ARE REQUIRED TO BE SHIPPED IN COMMERCIAL BOXCARS AND A PARTIAL LAYER RESULTS, THE PARTIAL LAYER OF LADING SHALL BE BRACED BY MEANS OF END BRACING AND/OR PARTIAL LAYER BRACING CONSTRUCTED IN ACCORDANCE WITH WR-52/100. SELECT THE TYPE OF BRACE TO COMPLY WITH THE WEIGHT OF THE UNITS TO BE RETAINED. THE CENTER GATE HEIGHT SHOULD BE ADJUSTED AS REQUIRED.
3. *THE LOADS AS SHOWN ARE BASED ON CARS WHICH HAVE 10 FT WIDE DOORWAY OPENINGS AND ARE EQUIPPED WITH CONVENTIONAL* SLIDING TYPE DOORS. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE APPLICABLE TO BOXCARS EQUIPPED WITH CONVENTIONAL SLIDING TYPE DOORS OTHER THAN 10 FT WIDE PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING.
4. THE DEPICTED PROCEDURES AND METHODS OF BLOCKING ARE ALSO APPLICABLE TO BOXCARS EQUIPPED WITH PLUG TYPE DOORS PROVIDING THE LENGTH OF PIECE 9 IS ADJUSTED AS REQUIRED TO SUIT THE PARTICULAR DOORWAY OPENING. DUNNAGE MATERIAL MUST NOT BE NAILED TO ANY PLUG DOOR, WHETHER MAIN OR AUXILIARY, EXCEPT WHEN THE CAR HAS A COMBINATION OF A CONVENTIONAL SLIDING TYPE DOOR AND A PLUG TYPE DOOR, AND AN ADEQUATE NAILING STRIP IS PROVIDED ON THE PLUG TYPE DOOR. IF LUMBER OF SUFFICIENT LENGTH TO SPAN PLUG DOORS IS NOT AVAILABLE, RANDOM LENGTH MATERIAL, DOUBLED AND SPLICED, BUT WITH JOINTS OF SPLICES OFFSET, MAY BE USED. STACKS WITH MORE THAN HALF OF THE UNIT LOAD IN THE DOORWAY AREA MUST BE UNITIZED WITH TWO LATERALLY APPLIED 1 1/4" STEEL STRAPS PER STACK, EACH TENSIONED AND SEALED WITH TWO DOUBLE CRIMPED SEALS AND A TOP SWAY BRACE CONSISTING OF A STRAPPING BOARD AND CROSS BRACE, TWO PER STACK (SEE DETAIL D) MUST BE INSTALLED BETWEEN THE ROWS OF THE DOORWAY STACKS. STRAP IS SECURED TO STRAPPING BOARD WITH TWO 1 1/4" STAPLES. DOORWAY PROTECTION, PIECES 11 AND 12 IS NOT REQUIRED WHEN PLUG DOOR EQUIPPED BOXCARS ARE USED, EXCEPT WHEN CAR HAS A COMBINATION OF PLUG DOOR AND CONVENTIONAL SLIDING DOOR. THEN DOORWAY PROTECTION IS REQUIRED FOR THE CONVENTIONAL DOOR. SECURELY CLOSE DOORS AND WIRE TOGETHER WITH A STRONG FLEXIBLE STEEL WIRE INSERTED THROUGH THE HOLES IN THE DOOR LATCH ASSEMBLY ONE OR MORE TIMES AND THE WIRE ENDS TWISTED TOGETHER.



LOADING PLAN

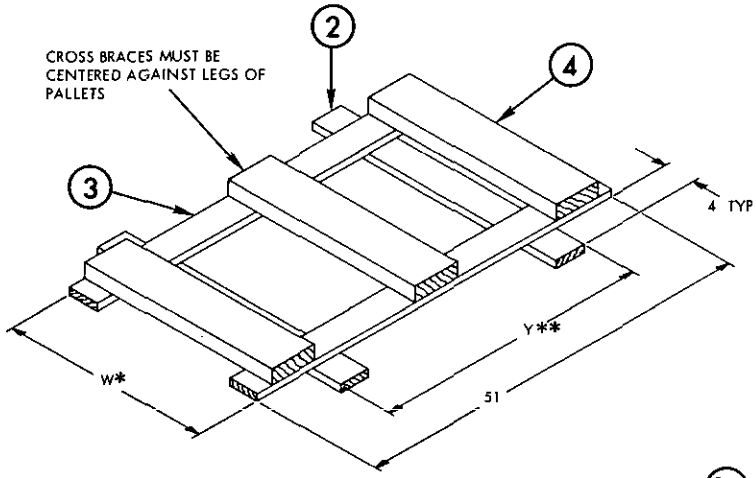
PIECE NO:	DESCRIPTION	SIZE	NO. OF PIECES REQ'D	NAIL TO	NUMBER	SIZE
					NAILS	
12	HORIZONTAL DOORWAY MEMBER	2 x 4 x DOOR WIDTH	8	11	2 PER JOINT	10d
11	VERTICAL DOORWAY MEMBER	2 x 3 x 96	4	DOOR POST	2 PER FOOT	20d
10	HOLD DOWN CLEAT	2 x 6 x 60	2	9	2 PER FOOT	10d
9	GATE HOLD DOWN	2 x 6 x 14 FT	2	CAR WALL	5 EACH END	10d
8	STRUT	4 x 4 x WEDGE FIT***	16	6	2 PER JOINT	16d
7	CENTER GATE STRUT CLEAT	2 x 4 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
6	CENTER GATE VERTICAL	2 x 6 x 90	8	SEE 5	-	-
5	CENTER GATE HORIZONTAL	2 x 6 x CAR WIDTH - 1	8	6	3 PER JOINT	10d
4	CROSS BRACE	2 x 6 x W*	33	SEE 3	-	-
3	SWAY BRACE LONGITUDINAL	1 x 4 x 51	22	2, 4	SEE NOTE**	
2	SWAY BRACE SUPPORT	1 x 4 x (W + 8)*	22	SEE 3	-	-
1	SLEEPER	2 x 4 x 51	22	CAR FLOOR	2 PER FOOT	16d
LIST OF MATERIALS & NAILING DATA						

*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**PIECE 3 NAILS TO PIECE 4 WITH 10d NAILS, CLINCHED, 3 PER JOINT, AND TO PIECE 2 WITH 6d NAILS, 1 PER JOINT.

***2 x 6 STRUTS DOUBLED AND *LAMINATED WITH 10d NAILS* MAY BE SUBSTITUTED IN PLACE OF 4 x 4's.

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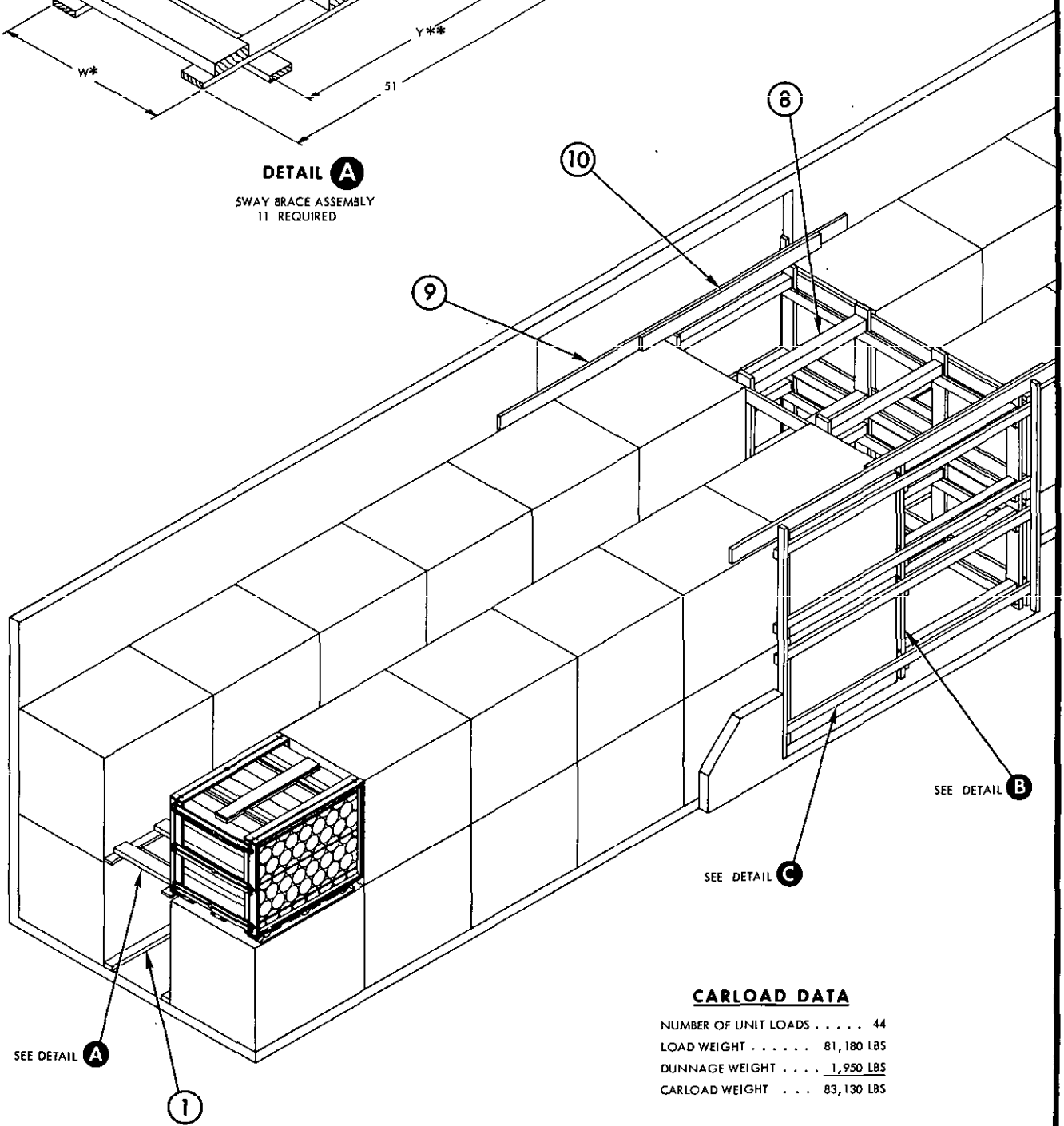
*W = DISTANCE BETWEEN LEGS OF PALLETS ACROSS THE CAR.

**Y = DISTANCE BETWEEN INSIDE OF OUTER LEGS OF PALLET.

INSERT SWAY BRACE SUPPORTS IN FORK POCKETS AND NAIL SWAY BRACE FRAME TO SUPPORTS.

DETAIL A

SWAY BRACE ASSEMBLY
11 REQUIRED

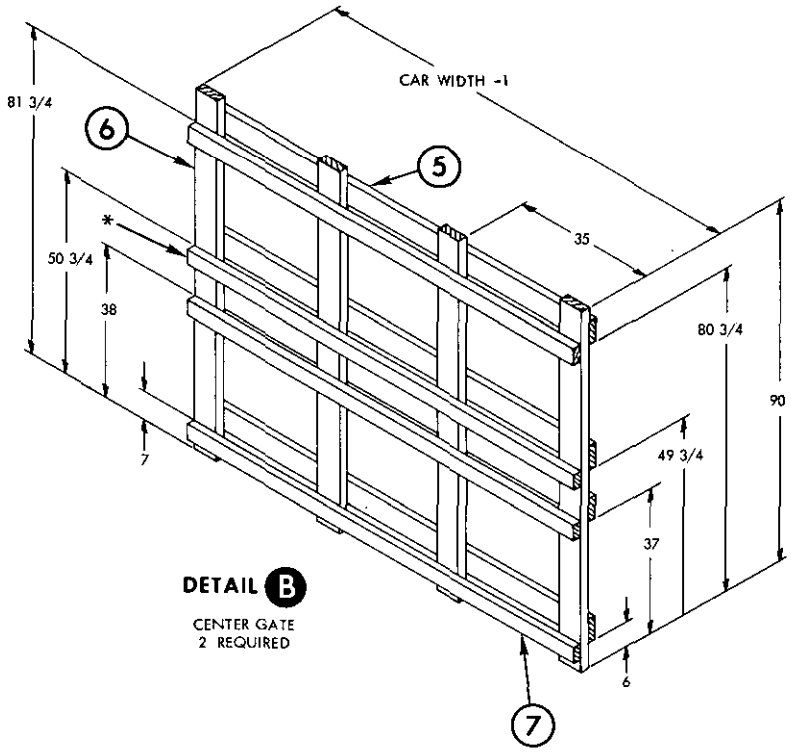


CARLOAD DATA

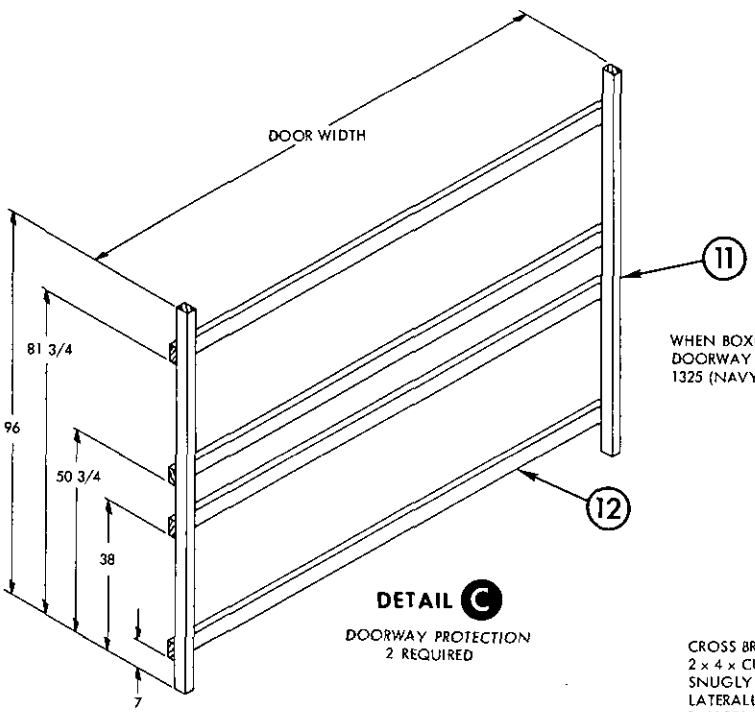
NUMBER OF UNIT LOADS	44
LOAD WEIGHT	81,180 LBS
DUNNAGE WEIGHT	1,950 LBS
CARLOAD WEIGHT	83,130 LBS

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*POSITION AFTER STRUTS HAVE BEEN INSTALLED ON LOWER STRUT CLEAT (REQUIRED ONE GATE ONLY)

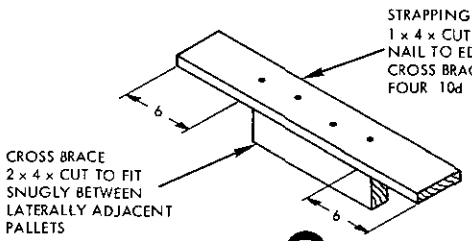


DETAIL B
CENTER GATE
2 REQUIRED



DETAIL C
DOORWAY PROTECTION
2 REQUIRED

WHEN BOXCAR HAS STEEL DOOR POSTS, DOORWAY PROTECTION AS PER MIL-STD-1325 (NAVY) PAGE 46, FIG. 21 IS REQUIRED.



DETAIL D
TOP SWAY BRACE

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